

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 17: Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Cyclic quadrilaterals

1. A, E, C and B lie in order on a circle. Chords AC and BC are equal, BC is produced to D , and $\angle ACD = 120^\circ$. Find $\angle AEC$. (a) 30° (b) 60° (c) 90° (d) 120° **ICSE 2025, 1 mark**

SOLUTION

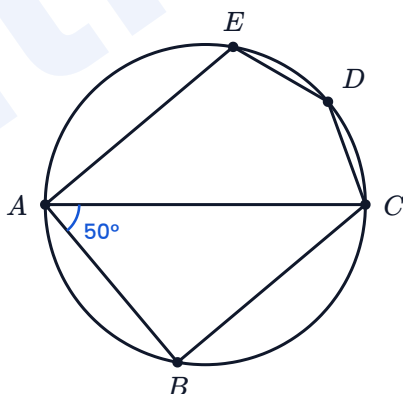
$\angle ACB = 180^\circ - 120^\circ = 60^\circ$ (linear pair).

$AC = BC$, so $\angle ABC = \angle BAC = \frac{180^\circ - 60^\circ}{2} = 60^\circ$.

$ABCE$ is a cyclic quadrilateral, so $\angle AEC = 180^\circ - \angle ABC = 120^\circ$.

Answer: (d)

2. $ABCDE$ is a pentagon inscribed in a circle. AC is a diameter, $BC \parallel AE$ and $\angle BAC = 50^\circ$. Find (a) $\angle ACB$ (b) $\angle EDC$ (c) $\angle BEC$. Then prove that BE is also a diameter. **ICSE 2019, 4 marks**



SOLUTION

(a) $\angle ABC = 90^\circ$ (angle in a semicircle), so $\angle ACB = 90^\circ - 50^\circ = 40^\circ$.

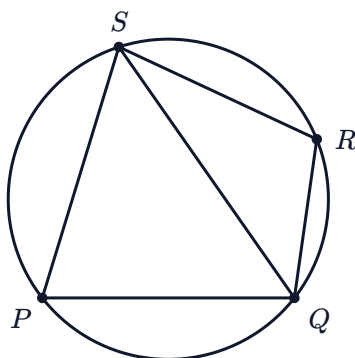
(b) $BC \parallel AE$, so $\angle EAC = \angle ACB = 40^\circ$ (alternate angles). $ACDE$ is a cyclic quadrilateral, so $\angle EDC = 180^\circ - 40^\circ = 140^\circ$.

(c) $\angle BEC = \angle BAC = 50^\circ$ (angles in the same segment).

$\angle EAB = \angle EAC + \angle CAB = 40^\circ + 50^\circ = 90^\circ$. A chord that subtends a right angle at the circumference is a diameter.

Answer: (a) 40° (b) 140° (c) 50° ; since $\angle EAB = 90^\circ$, BE is a diameter

3. $PQRS$ is a cyclic quadrilateral with $\angle QPS = 73^\circ$, $\angle PQS = 55^\circ$ and $\angle PSR = 82^\circ$. Find (a) $\angle QRS$ (b) $\angle RQS$ (c) $\angle PRQ$. **ICSE 2018, 4 marks**

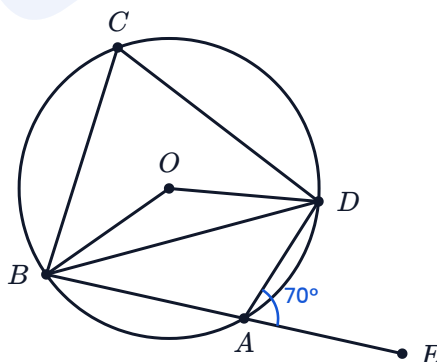


SOLUTION

- (a) Opposite angles of a cyclic quadrilateral add up to 180° : $\angle QRS = 180^\circ - 73^\circ = 107^\circ$.
 (b) In $\triangle PQS$: $\angle PSQ = 180^\circ - 73^\circ - 55^\circ = 52^\circ$, so $\angle QSR = 82^\circ - 52^\circ = 30^\circ$. In $\triangle QRS$: $\angle RQS = 180^\circ - 107^\circ - 30^\circ = 43^\circ$.
 (c) $\angle PRQ = \angle PSQ = 52^\circ$ (angles in the same segment).

Answer: (a) 107° (b) 43° (c) 52°

4. $ABCD$ is a cyclic quadrilateral in a circle with centre O . BA is produced to E and $\angle DAE = 70^\circ$. Find (a) $\angle BCD$ (b) $\angle BOD$ (c) $\angle OBD$. **ICSE 2017, 4 marks**



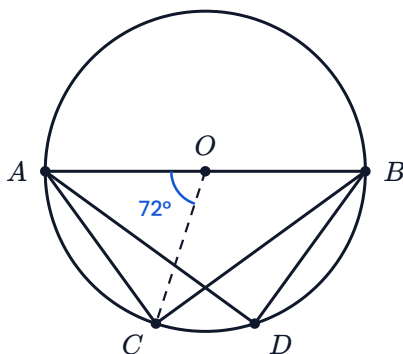
SOLUTION

- (a) The exterior angle of a cyclic quadrilateral equals the interior opposite angle: $\angle BCD = \angle DAE = 70^\circ$.
 (b) The angle at the centre is twice the angle at the circumference: $\angle BOD = 2 \times 70^\circ = 140^\circ$.
 (c) $OB = OD$ (radii), so $\angle OBD = \frac{180^\circ - 140^\circ}{2} = 20^\circ$.

Answer: (a) 70° (b) 140° (c) 20°

2 Angle at the centre and at the circumference

5. AB is a diameter of a circle with centre O . C and D are points on the circle on the same side of AB , with $AC = BD$ and $\angle AOC = 72^\circ$. Find (a) $\angle ABC$ (b) $\angle BAD$ (c) $\angle ABD$. **ICSE 2020, 3 marks**



SOLUTION

(a) The angle at the circumference is half the angle at the centre on the same arc: $\angle ABC = \frac{72^\circ}{2} = 36^\circ$.

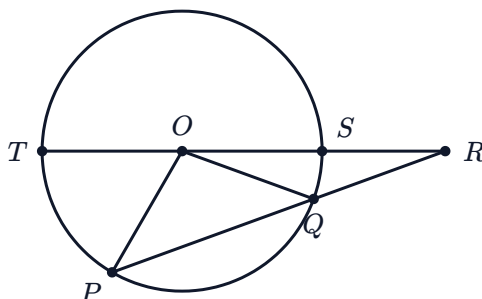
(b) Equal chords subtend equal angles at the centre, so $\angle BOD = 72^\circ$ and $\angle BAD = 36^\circ$.

(c) $\angle ADB = 90^\circ$ (angle in a semicircle), so $\angle ABD = 90^\circ - 36^\circ = 54^\circ$.

Answer: (a) 36° (b) 36° (c) 54°

3 Arcs, chords and angles

6. O is the centre of a circle. A line through O meets the circle at T and S and is produced to R . A line from R cuts the circle at Q and P , with Q nearer to R . $QR = OP$ and $\angle ORP = 20^\circ$. Find $x = \angle TOP$. **ICSE 2018, 3 marks**



SOLUTION

$OQ = OP$ (radii) and $QR = OP$, so $OQ = QR$. In $\triangle OQR$: $\angle QOR = \angle QRO = 20^\circ$.

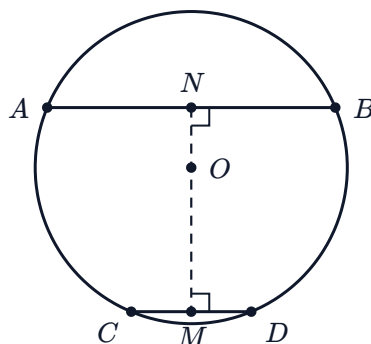
Exterior angle of $\triangle OQR$: $\angle OQP = 20^\circ + 20^\circ = 40^\circ$.

$OP = OQ$, so $\angle OPQ = \angle OQP = 40^\circ$.

x is an exterior angle of $\triangle OPR$: $x = \angle OPR + \angle ORP = 40^\circ + 20^\circ$.

Answer: $x = 60^\circ$

7. AB and CD are parallel chords on opposite sides of the centre of a circle of radius 13 cm. $AB = 24$ cm and $CD = 10$ cm. Find the distance between the chords. **ICSE 2017, 3 marks**



SOLUTION

Draw $ON \perp AB$ and $OM \perp CD$. The perpendicular from the centre bisects a chord, so $AN = 12$ cm and $CM = 5$ cm.

$ON = \sqrt{13^2 - 12^2} = 5$ cm and $OM = \sqrt{13^2 - 5^2} = 12$ cm

The chords are on opposite sides of the centre, so the distance is $ON + OM$.

Answer: 17 cm

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